*μ*LHC Physics

Owing to its high center-of-mass energy (500 GeV or 1 TeV) and high integrated luminosity (10^{33} cm^{-2} s^{-1} or 10^{34} cm^{-2} s^{-1}), the μLHC aims to make significant contributions to particle physics in terms of both precision measurements and energy-frontier exploration. The physics potential of the μLHC offers unique opportunities, particularly for advancing our understanding of the foundations of Quantum Chromodynamics (QCD), probing the properties of the Higgs boson, and searching for new physics Beyond the Standard Model (BSM).

1. *μ*LHC as a QCD Explorer

Despite the great success of the Standard Model, several fundamental phenomena of Quantum Chromodynamics that govern the strong interaction, which accounts for approximately 98% of the mass of the visible Universe, have not yet been experimentally established. In particular, essential QCD processes such as quark–gluon interactions, confinement, and hadronization remain incompletely understood. A comprehensive understanding of the fundamental dynamics of QCD and the internal structure of the proton critically depends on the kinematic reach accessible at lepton–hadron colliders. This reach is conventionally represented in the plane defined by the Bjorken-x and Q^2 variables, as shown in Figure 1.

By employing a point-like muon beam with no internal substructure, the μLHC provides a clean probe for direct access to quark and gluon dynamics within nucleons. This capability establishes one of the central scientific objectives of the μLHC as advancing our understanding of QCD in regimes that are inaccessible to existing experiments. The μLHC's reach in this critical region of the Q^2 phase space is illustrated in the Bjorken-x versus Q^2 plane shown in Figure 1.

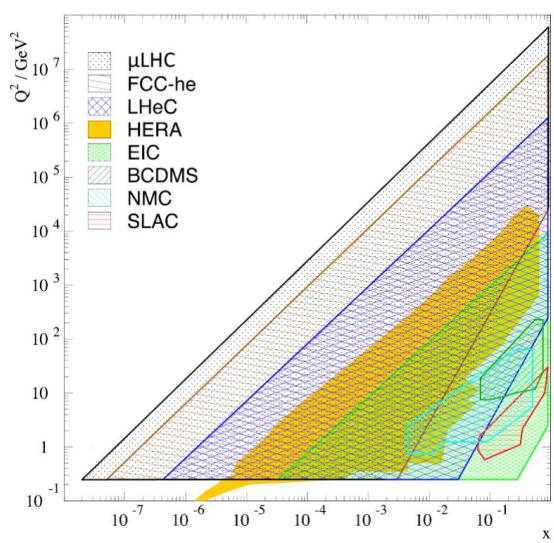

Figure 1. Kinematic plane of deep inelastic scattering [2]

The μLHC ($E_\mu = 1\,TeV$) has a center-of-mass energy that is approximately 16 and 4.5 times higher than those of HERA and the LHeC, respectively. Owing to this substantial increase in energy, the μLHC extends well beyond the projected capabilities of both HERA and the LHeC, as well as those of the FCC-eh, thereby providing a fundamentally broader coverage of the kinematic plane.

- At a fixed value of $x=10^{-4}$, the μLHC can reach momentum transfers of $Q^2 \approx 2800$ GeV^2. By comparison, the LHeC (ERL 50) reaches $Q^2 \approx 140$ GeV^2 at the same value of x, indicating that the μLHC provides approximately a factor of 20 higher Q^2 reach than the LHeC.
- At low momentum transfers of $Q^2=10$ GeV^2, the μLHC can probe record-low Bjorken-x values of approximately $x \approx 3.6\times10^{-7}$, whereas the LHeC (ERL 50) is limited to $x \approx 7.2\times10^{-6}$.

This extended kinematic coverage is vital for enabling the μLHC to test the fundamental principles of Quantum Chromodynamics, with particular emphasis on the investigation of high-density gluonic matter and parton saturation phenomena. The precise data to be provided by the μLHC will allow for an accurate determination of Parton Distribution Functions, which are essential inputs for analyses at future hadron colliders such as the HL-LHC, HE-LHC, and FCC or SppC.

2. *μ*LHC as a Higgs Factory

At lepton–proton colliders, Higgs boson production proceeds predominantly via the Vector Boson Fusion mechanism. The clean experimental environment provided by μp collisions offers an ideal setting for precision studies of the Higgs boson with high statistical accuracy.

Owing to the combination of its high integrated luminosity potential and high center-of-mass energy, the μLHC achieves more than an order-of-magnitude increase in the Higgs boson production rate compared to the planned LHeC colliders. This enhancement is clearly illustrated in Figure 2, which shows the dependence of the W- and Z-boson fusion cross sections (σ) on the center-of-mass energy ($\sqrt{s}$).

During event generation, Monte Carlo generators such as MadGraph5 employ generator-level cuts. In these calculations, the relevant selection criteria are chosen to be flexible, taking into account the acceptance regions and physics potential of prospective μLHC and LHeC detectors. Since MG5 applies these constraints directly within the matrix-element phase-space integration, the resulting cross-section values correspond to cut-dependent fiducial cross sections rather than fully inclusive theoretical cross sections.

The use of flexible cuts ensures that potentially relevant low-momentum signatures are not discarded at an early stage of the offline analysis. Furthermore, in order to maximize the coverage of the forward phase-space region, where processes are significantly boosted toward the proton direction due to beam asymmetry, the rapidity constraints are

extended. The generator-level selections are defined as: $bwcutoff = 15$, $\Delta R(l,j) > 0.4$, $p_{T,j} > 10\ GeV$, $|\eta_j| < 7$, $p_{T,l} > 5\ GeV, |\eta_l| < 4$.

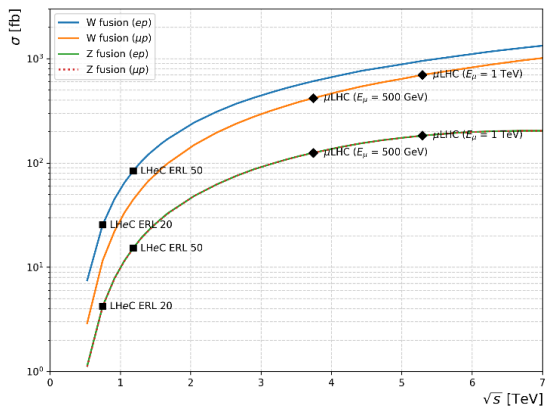

Figure 2. Higgs boson production cross sections at the μLHC and the LHeC as a function of the center-of-mass energy.

The values of the cross-sections at the relevant center-of-mass energies are based on the theoretical cross sections used to compute the annual production yields reported in Table 1. As shown in the table, for the μLHC operating at $\sqrt{s}$=5.29 TeV with an integrated luminosity of 100 fb^{-1} per year, the expected annual Higgs yields are estimated to be 69800 events via W-boson fusion and 18300 events via Z-boson fusion.

Table 1. Annual Higgs boson production via Vector Boson Fusion at the LHeC and the μLHC.

Collider	$\sqrt{s}\ [TeV]$	$L[cm^{-2}s^{-1}]$	$L_{int}[fb^{-1}/\text{year}]$	$\sigma_{WWH}\ [pb]$	$N_{Wfusion}/\text{year}$	$\sigma_{ZZH}\ [pb]$	$N_{Zfusion}/\text{year}$
LHeC (ERL 20)	0.748	10^{33} (10^{34})	10 (100)	0.025	250 (2500)	0.004	40 (400)
LHeC (ERL 50)	1.18	10^{34} (10^{35})	100 (1000)	0.084	8400 (84000)	0.015	1500 (15000)
μLHC (E_μ = 500 GeV)	3.7	10^{33} (10^{34})	10 (100)	0.418	4180 (41800)	0.125	1250 (12500)
μLHC (E_μ = 1 TeV)	5.3	10^{33} (10^{34})	10 (100)	0.698	6980 (69800)	0.183	1830 (18300)

3. Top Production

μLHC can shed light on new physics through precision measurements of top-quark couplings and the investigation of rare flavor-changing processes.

Table 2 presents the production cross sections for single-top and $t\bar{t}$ processes at the μLHC. Among the top-quark production channels considered, the dominant single-top process, $\mu^+\ p \rightarrow \bar{\nu}_\mu\ t\ X$, reaches a cross section as large as 31.22 pb. This sizeable cross section enables the μLHC to observe top-quark production events with high statistical

precision. When combined with the projected integrated luminosity of the μLHC, this high production rate allows for detailed studies of top-quark production with excellent statistical accuracy.

The μLHC thus offers significant potential to test the Standard Model through precision measurements of top-quark couplings, which form a cornerstone of SM analyses, as well as through investigations of rare flavor-changing processes. Such precision studies provide a robust framework for detailed explorations of electroweak interactions and potential effects of new physics.

Table 2. Cross sections for single-top and $t\bar{t}$ production processes at the μLHC.

Process	σ **[pb]** $E_\mu = 500\ GeV\ (1\ TeV)$
$\mu^+ p \rightarrow \bar{\nu}_\mu\, t\, X$	16.76 (31.22)
$\mu^+ p \rightarrow \bar{\nu}_\mu\, \bar{t}\, X$	$2.40\ (6.91) \times 10^{-7}$
$\mu^+ p \rightarrow \mu^+\, t\, X$	0.48 (0.91)
$\mu^+ p \rightarrow \mu^+ \bar{t}\, X$	0.48 (0.91)
$\mu^+ p \rightarrow \mu^+\, t\, \bar{t}$	0.48 (0.91)
$\mu^+ p \rightarrow \mu^+\, t\, \bar{t}\, X$	0.46 (0.96)
$\mu^+ p \rightarrow \bar{\nu}_\mu\, t\, \bar{t}\, X$	$4.56\ (9.49) \times 10^{-4}$

4. BSM

Despite the great success of the Standard Model, a number of experimental and theoretical tensions point toward the existence of new physics Beyond the Standard Model. One of the most prominent of these is the anomalous magnetic moment of the muon, $(g-2)_\mu$, which has exhibited a longstanding discrepancy between experimental measurements and Standard Model predictions. This anomaly may indicate the presence of new particles or interactions that couple preferentially to second-generation leptons. In addition, anomalies have been observed in heavy hadron decays, including possible violations of lepton flavor universality and deviations in flavor-changing neutral current processes, providing further evidence that new dynamics may be active in the flavor sector. Taken together, these experimental tensions and anomalies make it essential for high-energy colliders such as the μLHC to probe such BSM effects with exceptional precision.

One of the most significant advantages of the μLHC is its ability to access distinctive processes such as resonant (single) production and contact interactions, which are difficult or impossible to probe at other collider facilities.

The principal Beyond the Standard Model phenomena expected to be investigated at the μLHC include:

- **Leptoquarks:** Hypothetical particles that couple simultaneously to leptons and quarks.

- **Color-octet muons (μ8):** Strongly interacting partners of the muon, commonly referred to as leptogluons.
- **Excited muons and excited muon neutrinos.**
- **Squarks (Supersymmetry):** In particular, squarks arising in supersymmetric models with R-parity violation.
- **Contact interactions:** Effective four-fermion interactions at low energies induced by the exchange of new heavy states.

In addition to providing direct access to these particles, the μLHC can achieve indirect sensitivity to heavy BSM states through precise measurements of production cross sections and angular distributions.

Case Study: Color-Octet Muons (μ8)

The color-octet muon (μ8), also referred to as a leptogluon, provides an excellent example of the unique potential of the μLHC to explore new physics associated with the muon. Color-octet muons share the same electromagnetic and weak quantum numbers as the Standard Model muon, while carrying a QCD color charge, similar to quarks and gluons. This hybrid structure makes them an ideal probe for investigating possible compositeness of leptons.

At hadron colliders such as the LHC, μ8 particles are typically produced in pairs via gluon–gluon fusion, $p\,p \to \mu 8^{+} + \mu 8^{-}$. In contrast, at a μp collider, the direct interaction between the incoming muon (μ^{+}) and a gluon inside the proton allows for resonant single production, $\mu^{+}\,p \to \mu 8^{+}$. This resonant channel exhibits a significantly larger production cross section compared to pair production. The results of a dedicated study on μ8 production are presented in Figure 3 [2].

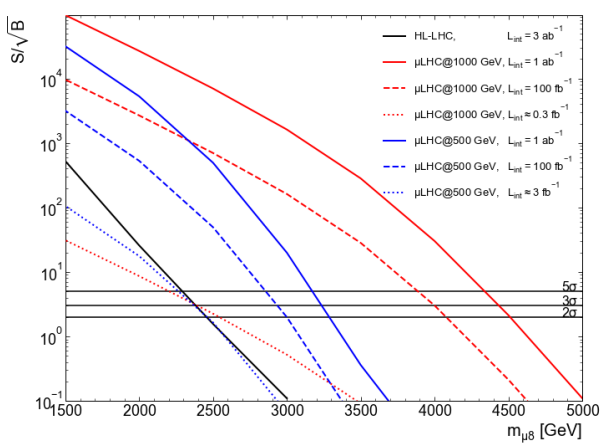

Figure 3. Statistical significance as a function of the μ8 mass.

As can be seen from the figure, the 5σ discovery mass reach at the HL-LHC, assuming an integrated luminosity of 3 ab^{-1}, is estimated to be approximately 2.3 TeV. In contrast, at the μLHC the corresponding discovery reach is projected to be around 4.3 TeV for an integrated luminosity of 1 ab^{-1}. Even more strikingly, the μLHC surpasses the discovery reach of the HL-LHC with an integrated luminosity as low as 300 nb^{-1}.

These results demonstrate that the μLHC constitutes a critical instrument for the future of high-energy physics and is expected to play a unique role, particularly in searches for new physics associated with the muon.

REFERENCES

1. Cheung, K., & Wang, Z. S. (2021). Physics potential of a muon-proton collider. Physical Review D, 103(11), 116009.
2. Akturk, D., Canbay, A. C., Dagistanli, H., Dagli, B., Kaya, U., Ketenoglu, B., Kilic, A., Kocak, F., Ozturk, A., Sultansoy, S., Tapan, I., Zimmermann, F. (2025). μLHC: Antimuon Ring and HL-LHC based μ+p Collider, https://doi.org/10.48550/arXiv.2506.15445.